

Question 1

Given that,

Population mean, $\mu = 120$ mmHg

Population SD, $\sigma = 15$ mmHg

Sample size, $n = 40$

Sample mean, $\bar{x} = 125$ mmHg

Significance level, $\alpha = 0.05$

Hypotheses:

- $H_0: \mu = 120$
- $H_1: \mu \neq 120$ (two-tailed test)

Test statistic:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{125 - 120}{\frac{15}{\sqrt{40}}} = \frac{5}{2.37} = 2.11$$

Critical value:

At $\alpha = 0.05$ (two-tailed), $Z_{0.025} = \pm 1.96$

Decision:

Since, $2.11 > 1.96 \rightarrow$ Reject H_0

Conclusion:

There is significant evidence that students' average blood pressure is different from the general population.

Question 2

Given,

$$\mu = 98.6^{\circ}\text{F}$$

$$\bar{x} = 99.1^{\circ}\text{F}$$

$$s = 0.5^{\circ}\text{F}$$

$$n = 10$$

$$\alpha = 0.05$$

Hypotheses:

- $H_0: \mu = 98.6$
- $H_1: \mu \neq 98.6$

Test statistic:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{99.1 - 98.6}{\frac{0.5}{\sqrt{10}}} = \frac{0.5}{0.158} = 3.16$$

Degrees of freedom:

$$df = 9$$

Critical value:

$$t_{0.025,9} \approx \pm 2.262$$

Decision:

Since, $3.16 > 2.262 \rightarrow \text{Reject } H_0$

Conclusion:

The drug raises body temperature significantly; it's not due to random chance.

Question 4

Given,

Species	$\bar{x}$	σ	n
A	75	8	60
B	72	7	55

Hypotheses:

- $H_0: \mu_1 = \mu_2$
- $H_1: \mu_1 \neq \mu_2$

Test statistic:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{75 - 72}{\sqrt{\frac{64}{60} + \frac{49}{55}}} = 2.14$$

Critical value:

- $Z_{0.025} = \pm 1.96$

Decision:

- $|2.14| > 1.96 \rightarrow \text{Reject } H_0$

Conclusion:

There is a significant difference in average heart rates between the two species.

Question 5

Given,

$$n = 10$$

$$p = 0.3$$

$$x = 4$$

$$\text{We know, } P(X=4) = {}^{10}C_4(0.3)^4 \times (1-0.3)^{10-4} = 210 \times 0.0081 \times 0.1176 = 0.20$$

About 20% chance that exactly 4 out of 10 individuals carry the gene

Question 6

Given,

$$\text{Mean, } \lambda = 3$$

$$x = 5$$

Now, Probability that exactly 5 colonies will form on a randomly selected dish is,

$$P(X=5) = \frac{e^{-3} \times 3^5}{5!} = 0.101$$

About 10% chance that exactly 5 colonies form on a dish.

Question 7

Given,

$$\mu = 37.5^\circ\text{C}$$

$$\sigma = 0.5^\circ\text{C}$$

Here, Probability that a randomly selected rat has a body temperature between 37°C and 38°C is,

$$\begin{aligned} P(37 < x < 38) &= P\left(\frac{37-37.5}{0.5} < \frac{x-\mu}{\sigma} < \frac{38-37.5}{0.5}\right) \\ &= P(-1 < Z < 1) \\ &= F(1) - F(-1) \\ &= 0.6826 \end{aligned}$$

Around 68% of rats have a body temperature between 37°C and 38°C .